

Vaibhav Garg

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience in machine learning, NLP and LLM-driven applications. Built and deployed AI solutions using Python, Scikit-learn, Flask, LangChain and Hugging Face, including loan approval prediction and AI travel concierge projects. Proficient in prompt engineering, RAG pipelines and model evaluation, seeking to support prototyping and deployment of enterprise AI systems.

EDUCATION

Maharaja Agrasen Institute of Technology

Aug 2023 - May 2027

Bachelor of Technology, Computer Science and Engineering (CSE)

• GPA: 9.26/10

• **Coursework:** Data Structures and Algorithms (DSA), Object-Oriented Programming (OOP), Database Management System (DBMS), Statistics, Probability

Saint Giri Senior Secondary School

2022 - 2023

Senior Secondary Education (XII)

• GPA: 92.8%

WORK EXPERIENCE

Edunet Foundation

Jun 2025 - Jul 2025

Artificial Intelligence & Machine Learning Intern

Remote

- Developed employee salary predictive model using Python and Gradient Boosting on 48,842 records with 15 features, delivering forecasts to inform compensation strategy.
- Performed data engineering tasks including data cleaning, outlier removal, and feature engineering in pandas, increasing data quality to 34,968 high-quality samples.
- Compared and evaluated five machine learning models (Logistic Regression, Random Forest, SVM, KNN, Gradient Boosting) to identify the optimal algorithm for salary prediction.
- Achieved a best accuracy of 84.56% using Gradient Boosting and packaged the optimized model for deployment, supporting product innovation.

PROJECTS

AI Travel Concierge - Airbnb [Source Code](#)

Jun 2026 - Aug 2026

- Built a RAG-powered conversational search engine over 100+ Airbnb-style listings using all-MiniLM-L6-v2 embeddings and ChromaDB vector store.
- Designed semantic retrieval pipeline that maps natural language vibe queries to top-5 most relevant listings - reducing search friction vs traditional filters
- Integrated Gemini API (gemini-2.5-flash) as conversational layer with grounded responses using retrieved listing context
- Supports multilingual input including Hinglish - built for Airbnb's 192-country user base.

NaturalCart | AI-Powered Grocery Shopping Platform [Source Code](#)

Jul 2026 - Aug 2026

- Developed a full-stack web application that parses natural language meal plans into aisle-organized, optimized grocery lists.
- Built a Retrieval-Augmented Generation pipeline using FAISS and SentenceTransformers (all-MiniLM-L6-v2) to perform semantic vector searches matching recipe ingredients to a catalog of 250+ products
- Integrated Google Gemini LLM to extract structured ingredient JSON schemas from unstructured text, applying dynamic filters for vegetarian, vegan, and gluten-free diets.
- Created a Smart Pack Optimizer to translate recipe quantity requirements into exact retail pack counts, reducing product waste and providing live checkout estimates.

Loan Approval Prediction System [Source Code](#)

Jun 2025 - Aug 2025

- Designed and implemented end-to-end loan approval prediction system using Python and Scikit-learn on 45,000+ loan records.
- Performed feature engineering and hyperparameter tuning using GridSearchCV, achieving 88.5% accuracy.
- Achieved 76% precision, 72% recall, and 74% F1-score, improving loan approval classification performance.
- Developed REST API using Flask and deployed scalable application on Render for real-time predictions.

Kindle Review Sentiment Analysis [Source Code](#)

Jan 2025 - May 2025

- Developed and deployed end-to-end NLP pipeline to classify sentiment from 11,000+ Amazon Kindle reviews
- Implemented Bag-of-Words and TF-IDF feature extraction techniques
- Trained Naive Bayes baseline model achieving 59% accuracy for sentiment classification
- Evaluated model performance using precision, recall, and F1-score metrics

TECHNICAL SKILLS

Programming Languages: Python, C++, SQL (MySQL), HTML, CSS, JavaScript

Data Analysis & Visualization: Data Analysis, Data Visualization, Pandas, NumPy, Matplotlib, Plotly

AI & Machine Learning: Machine Learning, Natural Language Processing (NLP), Prompt Engineering, Retrieval-Augmented Generation (RAG), Large Language Models (GPT, Claude, Gemini), Scikit-learn, NLTK, Predictive Modeling

Frameworks & Libraries: Flask, FastAPI, Streamlit, LangChain, Gradio, Hugging Face, OpenRouter, ChromaDB, BERT, FAISS

Developer Tools & Environments: Git, GitHub, Git Bash, VS Code, Cursor, Jupyter Notebook, Google Colab

CERTIFICATION

- Programming and Data Science (Foundation) – IIT Madras: 2023 – 2024
- Relevant Coursework: Python, Statistics, Probability, Computational Thinking